

(A copy of the answer in the 'Questionnaire' section)

Dear Dr. Stephenson-Jones Team,

I am writing to express my strong interest in the Research Assistant position in your lab focused on systems neuroscience. My academic and research background, combined with my deep passion for brain science, aligns closely with the requirements of this role. Throughout my academic journey, I have developed a profound interest in the complexities of brain mechanisms, particularly in how neural networks produce cognitive decisions and behaviours. My previous work in electrophysiological signal decoding and brain stem cell model observation has provided me with valuable insights into the challenges of both interpreting physiological signals and understanding cellular mechanisms. However, I have come to realise that a comprehensive understanding of the brain requires bridging the gap between molecular and behavioural levels, an endeavor that systems neuroscience is uniquely suited to address.

My educational background in Physics with Medical Physics has equipped me with a solid foundation in theoretical concepts, particularly in multi-modal biomedical imaging techniques, including light and advanced microscopy. During my master's studies, I further enriched this knowledge by focusing on neural data processing and analysis, where I applied computational approaches to gain a better understanding of the nervous system. Through simulations of computational models such as neuronal electrical properties, signal integration, reinforcement learning and other learning and memory models in MATLAB, I developed a deep understanding of how dynamic and statistical theories from physics and mathematics can be applied to computational neuroscience.

In addition to theoretical knowledge, I have honed practical skills in electrophysiological recordings and brain imaging data analysis through various research projects. For instance, I worked on decoding monkey movement trajectories from single and multiple unit recordings, which enhanced my ability to process large neural datasets and apply machine learning techniques. Moreover, I have refined my programming skills in Python, MATLAB, and C++ through engineering and biomedical applications, including processing data from EMG, ultrasound and advanced microscopy.

My research skills, developed across diverse lab environments, are highly transferable to this position. In my previous role as a research assistant, I independently conducted experiments to characterise an ultrasound system using a newly launched GUI. This involved designing experimental procedures, analysing data for software parameter calibration, and drafting end-user guidelines for a multidisciplinary audience. These tasks strengthened my abilities in data management, project development, troubleshooting, and effective communication with both team members and external collaborators. During my master's project on neurodegenerative disease model study, I

cultured cerebral organoids derived from human and mouse induced pluripotent stem cells to investigate early-stage neuronal development. This wet lab work required precise sample management, experiment scheduling, and strict protocol adherence, significantly improving my administrative skills. Additionally, I gained practical experience with techniques such as two-photon calcium imaging, which I believe would be directly applicable to the work in your lab. I am also highly motivated to gain hands-on experience in behavioural training with animal models. My commitment to delivering high-quality work through personal dedication and thoughtful collaboration, coupled with a strong sense of responsibility and safety in laboratory settings, would enable me to contribute effectively to your research team.

I have also developed my interpersonal skills through various work and volunteering experiences. These experiences have taught me the importance of contributing to an inclusive and respectful environment, as well as maintaining ethical standards and confidentiality in all aspects of my work. Leading a team of seven and serving as the presenter for a journal club and poster presentation have further enhanced my teamwork and collaboration abilities. I have demonstrated accountability and resourcefulness in delivering effective communication, organising multiple tasks, and making adjustments to ensure the success of any project.

My curiosity and motivation in systems neuroscience stem from not only from my academic journey but also from my personal life, where I approach cognitive feedback with a self-experimental mindset. Living in an era of information explosion and rapid AI advancements, it has become an inevitable challenge for individuals to process, select and make decisions amidst overwhelming external information. This drives my fascination with understanding how the human brain integrates the inputs and modulates internal states. The possibility of discovering interactive approaches and unraveling the mysteries of brain mechanisms is particularly compelling to me.

With my background in physics and neuroscience, I am excited about the potential to combine physics and mathematical models with experimental evidence in contributing to the field of theoretical neuroscience. I believe that the opportunity to work as a research assistant in your esteemed lab would be invaluable for advancing my knowledge and practical skills in neuroscience research. I am also keen to pursue a PhD in the near future and have a strong interest in the research topics at the Sainsbury Wellcome Centre. I hope to have the opportunity to pursue my PhD in this dynamic environment. I am eager to contribute to and learn from your team, and I am excited about the potential growth and development this position offers.

Yours sincerely,

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